NATIONAL CHEMISTRY CONTEST - CUBA 2020

“It always seems impossible until it’s done.”

Nelson Mandela

Name (s):______________________________ Last name:_____________________________

School:____

Municipality:______________________________ Province:______________________________

CODE:_______________________________ Exam: DAY 2. 12th GRADE

Instructions

- Write all your data clearly in the indicated place (names, surnames, school, municipality, province). Do not type anything in the CODE field.
- Check that your agenda is complete. This is the agenda for the second day of the exam, 12th grade. It is made up of 3 questions (question 10, 4 items; question 11, 7 items; question 12, 14 items) and a total of 6 pages.
- All necessary data, constants, conversions, and fundamental equations are included in the syllabus. You can use a scientific calculator.
- At the beginning of each question its value is indicated (out of a total of 100 points), as well as the value of each item (in marks).

Success!

Ministerio de Educación

República de Cuba

CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6{,}0221 \cdot 10^{23}\ mol^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gases contents	$R = 8{,}3145\ J \cdot K^{-1} \cdot mol^{-1}$	The third principle of thermodynamics	$Q = \Delta E + W$
Zeroth in the Celsius scale	$273{,}15\ K$	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485\ C \cdot mol^{-1}$	Nernst's equation	$E = E° - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{red}}{c_{ox}}$
Ionic product of water at 25 °C	$K_W = 1{,}00 \cdot 10^{-14}$	Arrhenius' equations	$k = A \cdot \exp\left(-\frac{E_A}{R \cdot T}\right)$
Standard conditions	$p° = 100\ kPa$ $c° = 1\ mol \cdot L^{-1}$	$\Delta G° = -R \cdot T \cdot \ln K° = -z \cdot F \cdot \Delta E°$	

$1\ J = 1\ C \cdot 1\ V = 1\ kPa \cdot 1\ L = 1\ W \cdot 1\ s$

$1\ C = 1\ A \cdot 1\ s$

PERIODIC TABLE OF THE ELEMENTS

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H 1.008																	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm -	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
89 Ac -	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np -	94 Pu -	95 Am -	96 Cm -	97 Bk -	98 Cf -	99 Es -	100 Fm -	101 Md -	102 No -	103 Lr -

Problem 10. Ionic equilibrium (10 points)

10.1	10.2	10.3	10.4	Total	Total
4	5	5	6	20	10

The following are examples of calculations that need to be performed frequently during the quantitative treatment of various electrolyte balances in aqueous solution.

10.1. What volume of 37% hydrochloric acid by mass (ρ_D = 1.19 g·ml–1) is needed to prepare 500 mL of a solution of approximate concentration 0.50 mol·L^{-1}?

10.2. Is a precipitate formed when an acid solution of pH 3 containing Pb^{2+} ions at a concentration of $5.0 \cdot 10^{-5}$ mol·L^{-1} is saturated with hydrogen sulfide gas? It is known that the total concentration of sulfide present under these conditions is close to 0.10 mol·L^{-1}.

$K_a(H_2S) = 1,0 \cdot 10^{-7}$; $K_a(HS^-) = 1,3 \cdot 10^{-13}$; $K_{ps}(PbS) = 2,5 \cdot 10^{-27}$

10.3. Calculate the solubility of silver sulfate in water in mol·L^{-1}. $K_{ps}(Ag_2SO_4) = 1,2 \cdot 10^{-5}$

10.4. Calculate the pH of the solution resulting from mixing 10 mL of 0.60 mol·L^{-1} hydrochloric acid with 20 mL of 0.50 mol·L^{-1} ammonia. $K_b(NH_3) = 1,8 \cdot 10^{-5}$

Problem 11. Cantharidin (19 points)

11.1	11.2	11.3	11.4	11.5	11.6	11.7	Total	Total
16	4	4	4	2	2	4	38	19

Cantharidin is a poisonous chemical compound produced by beetles of the Meloidae family. It was first isolated by the French chemist Pierre Jean Robiquet in 1812. It can be obtained starting from the Diels-Alder reaction between diene **A** and dienophile **B**, as shown below (Stork, G. *et al*, *J. Am. Chem. Soc.* **1951**, *73*, 4501; Stork, G. *et al*, *J. Am. Chem. Soc.* **1953**, *75*, 384-392).

11.1. Identify compounds **A**, **B**, **D**, **E**, **F**, **G**, **L** and **M**.

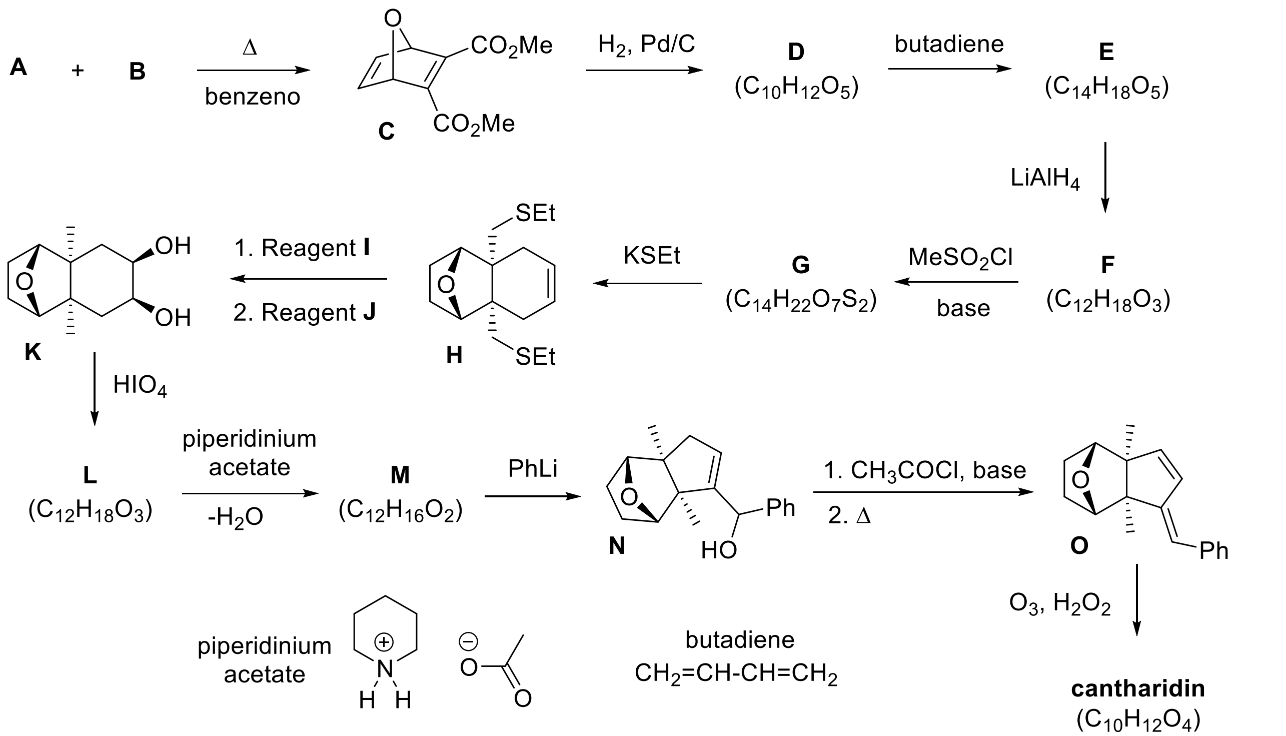
11.2. Draw the structure of cantharidin. It is known that its structure is symmetrical, it does not add bromine water and it reacts with amines and hydrazines.

11.3. State the balance between the s-cis and s-trans conformations around the central single bond of butadiene; Indicate which of the two forms is found in greater proportion. Use Newman projections to represent the conformers. Say which of them is the one that participates in the cycloaddition reaction.

11.4. Suggest the reagents used to transform **H** into **K**. Reagent **I** makes it possible to hydroxylate the alkene present in the tricycle **H**; while reagent **J** allows to reduce the C-S bond.

11.5. Label the stereogenic centers of compound **N**. How many stereoisomers are theoretically possible?

11.6. To use phenyllithium it is necessary to maintain an inert atmosphere. State one of the possible side reactions that can occur if this precaution is not taken into account.



11.7. Propose a way to obtain phenyllithium (PhLi) using benzene as the only organic starting product.

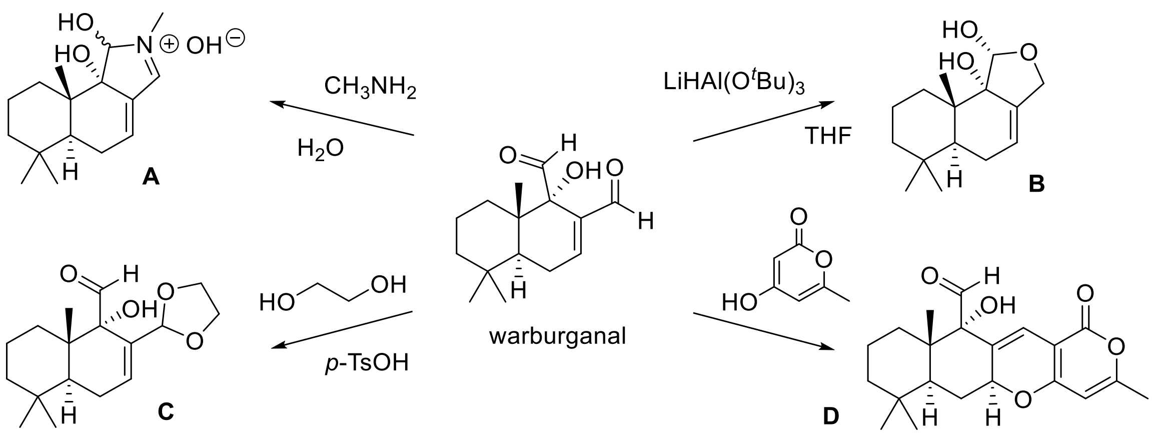
Problem 12. Thujopsene, warburganal and shikimic acid (21 points)

12.1	12.2	12.3	12.4	12.5	12.6	12.7	12.8
2	7	2	2	2	4	4	2
12.9	**12.10**	**12.11**	**12.12**	**12.13**	**12.14**	**Total**	**Total**
2	2	4	4	2	3	42	21

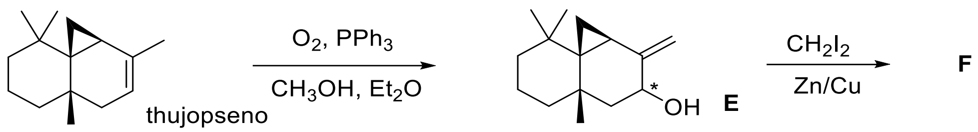
Semi-synthesis or partial synthesis consists of chemically obtaining compounds of interest starting from a natural product that has not been previously synthesized, but rather was extracted and purified from a microorganism using mixture separation techniques. It is used in industry and in research when it is a better alternative to a total synthesis, generally taking into account time and cost criteria. Warburganal, thujupsene and shikimic acid are three natural products that have been used in partial synthesis processes. Below are examples of chemical modifications that can be made to these substances.

12.1. In the structure of warburganal, point to the most reactive carbonyl group. How do you explain the difference in reactivity between like functional groups in the molecule?

12.2. Propose mechanisms for all the reactions represented in the schematic. p-TsOH: *para*-methylbenzenesulfonic acid.



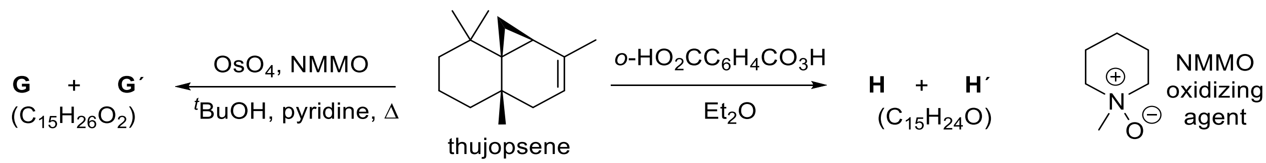
12.3. Thujupsene, in the presence of oxygen from the air and triphenylphosphine, is oxidized to allyl alcohol **E**. Propose a mechanism for the reaction described.



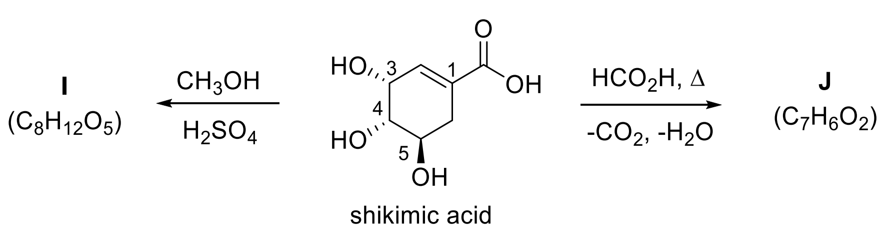
12.4. Which stereoisomer of alcohol **E** do you think is the major product in the above reaction? Represent the corresponding structure.

12.5. Suppose **E** (the isomer you found in 12.4) is reacted with the Simmons-Smith reagent. Represent the structure of the major product **F** that is obtained; note the stereochemistry.

Product of the presence of a double bond in its structure, thujopsene reacts with OsO_4 and o-peroxybenzoic acid, as shown below.



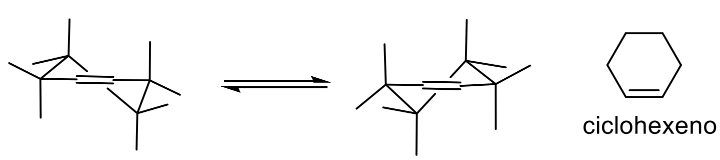
12.6. Represent the structures of **G** and **H**, and their respective stereoisomers **G´** and **H´**. It is known that **H** and **H´** are not the expected products of the reaction of thujopsene with a peracid: both have only four asymmetric carbons in their structure and react with 2,4-dinitrophenylhydrazine.



12.7. Represent the structures of **I** and **J**.

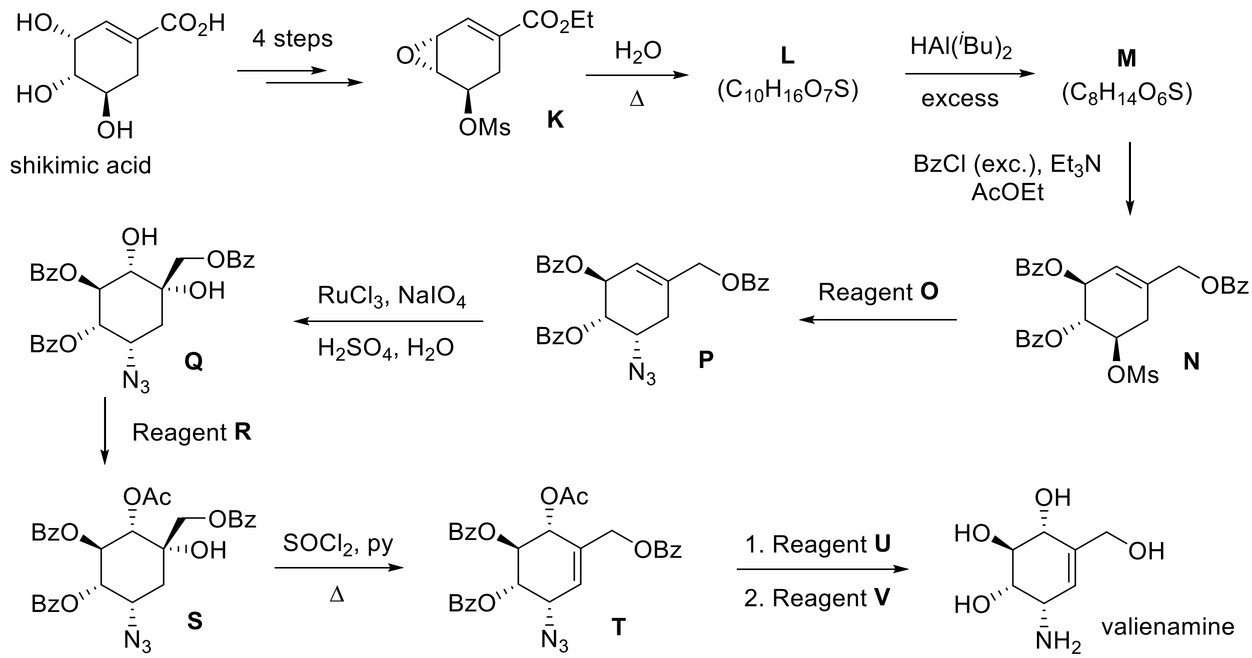
12.8. How could you selectively protect the hydroxyl groups that occupy positions 3 and 4 in the shikimic acid molecule?

12.9. Draw the most stable conformation for shikimic acid. Assume that the molecule is in the form of a semi-chair,



like cyclohexene. Show only the substituents that are not hydrogen atoms.

Shikimic acid can be used as a starting material for the synthesis of aminocyclitol valienamine. This compound is found as part of several pseudo-oligosaccharides such as the antidiabetic drug acarbose and the antibiotic validamycin.



12.10. Name the compound valienamine according to IUPAC nomenclature.

12.11. Represent the structures of **L** and **M**. Ms: methanesulfonyl or mesyl, $-SO_2CH_3$.

12.12. Suggest the possible reagents identified in the scheme as **O**, **R**, **U**, and **V**. It is known that **U** allows the hydrolysis of the acetyl (Ac) and benzoyl (Bz) groups, while **V** allows the azide group to be reduced to amino.

12.13. State a mechanism for the transformation of **S** into **T**.

In the above sequences, osmium(VIII) and ruthenium(VIII) oxides are used to dihydroxylate thujopsene and alkene **P**, respectively. As osmium and ruthenium compounds are very toxic and expensive, one way to carry out this transformation is to generate these reagents in situ by the action of oxidizing agents such as N-methylmorpholine oxide (NMMO) or potassium periodate.

12.14. Fit the chemical equation for the reaction for the in situ formation of ruthenium(VIII) oxide from ruthenium(III) chloride by the ion-electron method.

$$\mathrm{RuCl_3(ac) + IO_4^-(ac) \rightarrow RuO_4(ac) + IO_3^-(ac)}$$